WEEK 6 · LAB 4 · PROJECT STUDIO

Execute the Reproduction

Run your group's frozen M1 plan on its assigned study and build an evidence chain that another person can inspect.

The point of this studio

Every group is working on a different study, so there is no common repository, dataset or "correct answer" today. The common task is the process: follow the frozen M1 plan, execute the bounded reproduction, preserve raw evidence, handle deviations transparently, and reach a defensible current reproduction status.

1 · Re-open the frozen M1 contract

Before running anything substantial, read the M1 entry your group froze in Week 5 alongside the assigned study/project pack. Do not replace the RQ, sample, metric or decision rule because you now think a different route may give a better-looking result.

M1 item	Confirm before execution
RQ + target	The exact assigned RQ(s) and original result/claim you are reproducing.
Bounded sample	The frozen unit, source/version, size/range, selection rule and exclusions.
Method	The frozen inputs → procedure → output route, including material parameters/configuration.
Metric / result	What output you will calculate or observe and how it connects to the RQ.
Decision rule	The pre-declared rule for interpreting REPRODUCED / NOT REPRODUCED / INCONCLUSIVE.
Held-out boundary	Only the initial material currently released to your group. Do not use Event/held-out material.

No answer-chasing

A reproduction is useful even when it does not match the paper. Your job is to produce trustworthy evidence, not to keep changing the experiment until the original result appears.

2 · Prepare one traceable run

Set up the exact artefact/data/version required by M1 and make the execution route explicit. In your project repository or evidence location, preserve enough information for another person to understand what was run without relying on your memory.

Preserve	Minimum evidence
Input identity	Dataset/repository/release/artefact version and the bounded items actually used.
Environment	Relevant tool/language/library/version or configuration that materially affects the run.
Execution route	The command, script, notebook step or other procedure used to generate the result.
Raw evidence	Logs, CSV/JSON/text output, result files or other original machine/artefact output. Prefer raw evidence over screenshots where possible.
Derived result	The table, count, metric, comparison or other output calculated from that raw evidence.

3 · Prove the pipeline on a small valid case

Before launching the whole bounded sample, execute the method on one small representative item or the smallest valid portion of the frozen sample. This is a pipeline check: can the intended input travel through the intended procedure and produce the intended form of output? It is not permission to choose only the easiest case for the final result.

If the pilot exposes a problem

Fix ordinary execution faults where possible. If the fix changes a material M1 decision — sample, method, parameter, metric or interpretation rule — preserve the original M1 entry and record a deviation BEFORE the new run: what changed, why it was necessary, and how it may affect interpretation.

4 · Run the bounded reproduction

Now execute the frozen sample. Keep failed/awkward cases visible. Do not silently remove an item, change a threshold, tune a parameter or switch to a different metric because the current output is inconvenient.

Allowed	Not allowed
Repair a path, dependency or command error that does not change the empirical method.	Change the frozen sample after seeing which items give favourable results.
Repeat a run when the first execution is demonstrably invalid; preserve why it was invalid.	Drop failed cases without applying the frozen exclusion rule and recording them.
Make a necessary methodological deviation and record it before rerunning.	Change the metric, tolerance, parameter or decision rule after seeing the output without declaring a deviation.
Use GenAI to troubleshoot code/environment, then verify the proposed fix yourself.	Treat an AI explanation or generated number as empirical evidence.
Preserve each valid raw output used in the analysis.	Overwrite the only copy of raw evidence with a cleaned/edited result.
Stop and record a genuine blocker when a valid run cannot be produced.	Use held-out/Event material to rescue or enlarge the Week 6 result.

5 · Build the evidence chain before interpreting

Your result must be traceable backwards. Pick at least one result you plan to report and make sure you can point from the claim all the way back to the raw execution evidence that produced it.

FROZEN INPUT	RAW EXECUTION EVIDENCE	DERIVED RESULT / COMPARISON
The bounded cases/version selected by the M1 rule.	The actual logs/files/outputs produced by the reproduction procedure.	The metric/table/ordering/count used to compare with the original study.

6 · Compare with the original study

Only after you have a valid result should you apply the M1 decision rule. A different result is not automatically a mistake; a matching result is not automatically trustworthy. First decide whether the run itself is valid and sufficiently complete.

Current status	Use it when
REPRODUCED	A valid completed bounded run satisfies the pre-declared criterion for the original result/claim.
NOT REPRODUCED	A valid completed bounded run fails the pre-declared reproduction criterion.
INCONCLUSIVE / BLOCKED	Execution, data, environment or evidence prevents a valid comparison.
IN PROGRESS	The pipeline is valid but the bounded sample/result is not yet complete. This is a progress state, not a scientific outcome; complete it before M2.

7 · Coordinator reproduction checkpoint

Show the coordinator evidence from your own study. The coordinator does not need every group to use the same tool or output format; they need to see that your evidence chain is real, bounded and consistent with M1.

Show	What should be visible
Frozen M1	The RQ, bounded sample, method/metric and decision rule you committed to before the run.
Executable route	The actual command/script/notebook/procedure used for this reproduction.
Raw evidence	At least one genuine raw output from the current bounded reproduction.
Traceability	You can trace one reported/derived result backwards to its raw evidence and input.
Deviation record	Any material change from M1 is explicit, dated/reasoned and was not silently rewritten.
Current status	REPRODUCED / NOT REPRODUCED / INCONCLUSIVE-BLOCKED / IN PROGRESS is justified by the evidence.
Held-out boundary	No Event/held-out material has been consumed.

The coordinator may ask you to trace one result backwards

Be ready to start from one number/claim in your current result and show: where it came from → the raw output → the run/procedure → the exact input/case. If that chain breaks, the evidence is not ready yet.

8 · Leave Week 6 ready for M2

Before leaving, make sure the current work is preserved in your project repository/evidence location. Your group should know its current reproduction status, what evidence supports that status, what deviations exist, and the exact next action required before the M2 reproduction checkpoint. Do not consume Event material early.

AI rule

GenAI may help explain code, write support scripts or troubleshoot execution. You remain responsible for verifying every change. If an AI suggestion changes a material part of the frozen empirical method, treat it exactly like any other methodological deviation: record it before rerunning and explain its consequence.

Week 6 in one line

Run what you froze, preserve what actually happened, change the plan only transparently, and let the evidence — not the desired answer — determine the reproduction status.